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Avogadro's constant	$N_A = 6,0221 \cdot 10^{23} \text{ mol}^{-1}$	Ideal gas equation	$p \cdot V = n \cdot R \cdot T$
Gas' constant	$R = 8,3145 \text{ J} \cdot \text{K}^{-1} \cdot \text{mol}^{-1}$ $= 0,08205 \text{ atm} \cdot \text{L} \cdot \text{K}^{-1} \cdot \text{mol}^{-1}$	First Principle of Thermodynamics	$Q = \Delta U + W$
Zero degrees Celsius	273,15 K	Work	$W = p \cdot \Delta V = F \cdot d$
Faraday's constant	$F = 96485 \text{ C} \cdot \text{mol}^{-1}$	Gibbs Energy	$\Delta G = \Delta H - T \cdot \Delta S$
Ionic product of water at 25 °C	$K_W = 1,00 \cdot 10^{-14}$	Nernst's equation	$E = E^\circ - \frac{R \cdot T}{z \cdot F} \cdot \ln \left(\frac{c_{\text{red}}}{c_{\text{ox}}} \right)$
Arrhenius equation	$k = A \cdot \exp \left(-\frac{E_A}{R \cdot T} \right)$	$\Delta_r G^\circ = -R \cdot T \cdot \ln K^\circ = -z \cdot F \cdot \Delta E^\circ$	
First-order kinetics	$\ln c_t = \ln c_0 - k_r \cdot t$	Van't Hoff factor	$i = 1 + \alpha \cdot (v - 1)$
	$t_{1/2} = \frac{\ln 2}{k_r}$	Cryoscopy	$\Delta T_{\text{crio}} = i \cdot K_{\text{crio}} \cdot b$
Absorbance	$A_\lambda = -\log_{10} \left(\frac{I}{I_0} \right) = \varepsilon \cdot l \cdot c$	Drop in vapor pressure	$\Delta p = \frac{i \cdot x_s}{x_{\text{dis}} + \sum i \cdot x_s} \cdot p^\circ_{\text{dis}}$
Volume of a sphere	$V_{\text{sp}} = \frac{4}{3} \cdot \pi \cdot r^3$	Standard conditions	$p^\circ = 1 \text{ bar}$ $c^\circ = 1 \text{ mol} \cdot \text{L}^{-1}$
Area of a sphere	$A_{\text{sp}} = 4 \cdot \pi \cdot r^2$	Energy of a photon	$E = h \cdot \nu = h \cdot \frac{c}{\lambda}$
Area of a circle	$A_{\text{sp}} = \pi \cdot r^2$	Speed of light	$c = 2,998 \cdot 10^8 \text{ m} \cdot \text{s}^{-1}$
Long. of a circumf.	$l_c = 2 \cdot \pi \cdot r$	Planck's constant	$h = 6,626 \cdot 10^{-34} \text{ J} \cdot \text{s}$
Force of gravity	$F_g = m \cdot g$	Gravity on Earth	$g = 9,81 \text{ m} \cdot \text{s}^{-2}$
$1 \text{ J} = 1 \text{ C} \cdot \text{V} = 1 \text{ A} \cdot \text{s} \cdot \text{V} = 1 \text{ kPa} \cdot \text{L} = 1 \text{ Pa} \cdot \text{m}^3 = 1 \text{ N} \cdot \text{m} = 1 \text{ W} \cdot \text{s} = 0,2389 \text{ cal}$			
$1 \text{ N} = \text{kg} \cdot \text{m} \cdot \text{s}^{-2}$			
$1 \text{ atm} = 760 \text{ Torr} = 760 \text{ mm} = 101,325 \text{ kPa}$			
$1 \text{ bar} = 100 \text{ kPa}$			
$1 \text{ eV} = 96,485 \text{ kJ} \cdot \text{mol}^{-1}$			
$\pi = 3,1416$			
$1 \text{ pm} = 0,01 \text{ \AA} = 10^{-12} \text{ m}$			
$1 \text{ dm}^3 = 1 \text{ L}$			
$1 \text{ mL} = 1 \text{ cm}^3$			
$1 \text{ M} \equiv 1 \text{ mol/L}$			

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1	1 H 1.008																	2 He 4.003
2	3 Li 6.941	4 Be 9.012											5 B 10.811	6 C 12.011	7 N 14.007	8 O 15.999	9 F 18.998	10 Ne 20.180
3	11 Na 22.990	12 Mg 24.305											13 Al 26.982	14 Si 28.086	15 P 30.974	16 S 32.066	17 Cl 35.453	18 Ar 39.948
4	19 K 39.098	20 Ca 40.078	21 Sc 44.956	22 Ti 47.867	23 V 50.942	24 Cr 51.996	25 Mn 54.938	26 Fe 55.845	27 Co 58.933	28 Ni 58.693	29 Cu 63.546	30 Zn 65.39	31 Ga 69.723	32 Ge 72.61	33 As 74.922	34 Se 78.96	35 Br 79.904	36 Kr 83.80
5	37 Rb 85.468	38 Sr 87.62	39 Y 88.906	40 Zr 91.224	41 Nb 92.906	42 Mo 95.94	43 Tc 98.906	44 Ru 101.07	45 Rh 102.91	46 Pd 106.42	47 Ag 107.87	48 Cd 112.41	49 In 114.82	50 Sn 118.71	51 Sb 121.75	52 Te 127.60	53 I 126.905	54 Xe 131.29
6	55 Cs 132.91	56 Ba 137.33	57 La 138.91	* 72 Hf 178.49	73 Ta 180.95	74 W 183.84	75 Re 186.21	76 Os 190.23	77 Ir 192.22	78 Pt 195.08	79 Au 196.97	80 Hg 200.59	81 Tl 204.38	82 Pb 207.2	83 Bi 208.98	84 Po [209]	85 At [210]	86 Rn [222]
7	87 Fr [223]	88 Ra [226]	89 Ac [227]	** 104 Rf [265]	105 Db [268]	106 Sg [271]	107 Bh [270]	108 Hs [277]	109 Mt [276]	110 Ds [281]	111 Rg [280]	112 Cn [285]	113 Nh [284]	114 Fl [289]	115 Mc [288]	116 Lv [293]	117 Ts [294]	118 Og [294]

*	58 Ce 140.12	59 Pr 140.91	60 Nd 144.24	61 Pm [145]	62 Sm 150.36	63 Eu 151.96	64 Gd 157.25	65 Tb 158.93	66 Dy 162.50	67 Ho 164.93	68 Er 167.26	69 Tm 168.93	70 Yb 173.04	71 Lu 174.97
**	90 Th 232.038	91 Pa 231.036	92 U 238.029	93 Np [237]	94 Pu [242]	95 Am [243]	96 Cm [247]	97 Bk [247]	98 Cf [251]	99 Es [252]	100 Fm [257]	101 Md [258]	102 No [259]	103 Lr [262]

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Day 2. Grade 11th. Questions 14, 15 and 16.



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14.4. During the decomposition process $J \rightarrow F$ by heating four oxygenated binary compounds have been isolated, with 54.9, 56.5, 57.5% and 61.9% by mass of **X**, respectively, of which **F** is the one with the lowest oxygen content and is the only stable one. Calculate the formula for the three unstable compounds.

14.5. What are oxides and hydroxides, such as **E** and **C**, capable of reacting with acids and bases, called? Formulate two equations that illustrate dual behavior with one of these substances as example.

14.6. Tell the type of bond and particle that exists in the oxides **E**, **F** and **J**. Explain this variation.

14.7. What happens when ammonia is added little by little, until an excess is achieved, to a solution of cation **B**? Formulate the corresponding chemical reactions and tell what changes are observed.

14.8. How can **K** salt be obtained?

X is found naturally in two minerals, which also contain another metal in the 2+ oxidation state and oxygen. Both minerals can be described as equimolar mixtures (1:1) of two metal oxides. A mineral contains **X** in its most stable oxidation state and 24.9% of an unknown metal **Y**. The other mineral contains **X** in its maximum oxidation state and 64.1% of an unknown metal **Z**.

14.9. Determine which metals **Y** and **Z** are, as well as the empirical formulas for minerals.

Question 15. A salty question (12% of total; only 11th grade) Day 2

15.1	15.2	15.3	15.4	15.5	15.6	Total	Total
8	4	6	10	16	4	48	12

Commercial "low-sodium salts" often consist of mixtures of NaCl (< 30% by mass) and KCl. In these formulations, the element potassium is added to reduce dietary sodium intake. Now imagine that you are a chemical worker in a laboratory, and you must control the quality of a formulation using the following experimental procedure.

- 1- Dissolve 3,000 g of the commercial salt in distilled water, completing up to 250 mL in a volumetric flask.
- 2- Take a 50 mL aliquot of the previous solution and add an excess of 0.5 M silver nitrate until quantitative precipitation of the chloride ion is achieved. Filter the precipitate and wash with small portions of ice water. Dry the precipitate by slow heating until constant weight.

15.1. Suppose that during the analysis of a sample **M**, the mass obtained from AgCl precipitate was 1.157 g. Calculate the total chloride content (in % by mass) present in the sample.

Since the previous method has the disadvantage of losses due to precipitate handling, you now carry out the analysis using reverse titration by the Volhard method.

- 1- Take a 25 mL aliquot of the sample solution, add a 25 mL aliquot of 0.400 M AgNO₃ standard solution, and add a few drops of acidic iron(III) chloride indicator solution with nitric acid.
- 2- Titrate with a standard solution of potassium thiocyanate 0.100 M until the appearance of the carmine-red color of the [Fe(SCN)]²⁺ complex.

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15.2. Why is it necessary to add acid to the FeCl_3 solution? Write the relevant chemical equation.

15.3. Write down the equations of all the reactions that occurred during the second determination.

15.4. Suppose that during the analysis of the same M sample, the consumed volume of thiocyanate titrating reagent was 58.1 mL. Calculate the total chloride content (in % by mass) present in the sample.

15.5. Using the data from this second determination, determine the composition (in % by mass) of sample **M** and say whether it can be classified as "low-sodium".

15.6. Consider the following data for complex ion formation and solubility product constants. Explain the different affinity of the SCN^- and Cl^- anions to the Ag^+ and Fe^{3+} cations.

$K_{\text{ps}}(\text{AgCl}) = 1,8 \cdot 10^{-10}$ $K_{\text{ps}}(\text{AgSCN}) = 1,0 \cdot 10^{-12}$ $K_{\text{f}}[\text{Fe}(\text{SCN})]^{2+} = 890$ $K_{\text{f}}[\text{Fe}(\text{Cl})]^{2+} = 115$

Question 16. Phosphate determination (20% of total; only 11th grade) Day 2

16.1	16.2	16.3	16.4	16.5	16.6	16.7	16.8	Total	Total
4	12	6	12	10	8	6	22	80	20

Breakfast cereals are usually fortified with essential minerals. In this analysis, the goal is to determine the content of inorganic phosphorus (in the form of phosphate) using three different techniques: neutralization volumetry, spectrophotometry and gravimetry.

A 5.0000 g sample of a commercial cereal was pulverized and incinerated. The ashes were dissolved in a few drops of concentrated HCl , and distilled water was added. It was gently heated to remove excess volatile acid. After cooling, the resulting solution is transferred to a 250 mL volumetric flask, and distilled water is added to bring the volume to the mark (**M₀ Stock Solution**). In this step, all the phosphate was transformed into dissolved H_3PO_4 .

Part I. Initially, a 25.0 mL aliquot of the **M₀** solution was taken, and 10.0 mL of H_2SO_4 0.0130 M was added. This mixture was diluted with distilled water to a final volume of 100 mL in a volumetric flask (**M₁ Solution**). An aliquot of 25.0 mL of the new solution was titrated with 0.0151 M NaOH , using thymolphthalein as an indicator (turning interval 9.3-10.5). The average volume of the base to reach the endpoint of the titration was 8.5 mL. It is known that the pH of the final solution of the titration was close to 10. Consider that the only protolytically active component present in the solution obtained from the cereal is phosphate. $\text{pK}_a(\text{H}_3\text{PO}_4) = 2,15; 7,20; 12,32$ a 25 °C.

16.1. Which of the phosphoric acid species will be present in greater proportion at the beginning and at the end point of the titration?

16.2. Calculate the molar concentration of phosphate in solution **M₁**.

16.3. Determine the phosphorus content (in mg) in the initial cereal sample. If you were unable to determine the phosphate concentration in the **M₁** solution, use $1.0 \cdot 10^{-3}$ mol/L for the calculations.

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16.4. Calculate the pH of the **M₁** solution obtained before performing the titration. If you were unable to determine the phosphate concentration in the **M₁** solution, use $1.0 \cdot 10^{-3}$ mol/L for the calculations. Suppose sulfuric acid is strong in its two stages of dissociation.

Part II. The above method may be compromised by the presence of other acidic species in the sample. A selective alternative for phosphate quantification is spectrophotometric determination via formation of the molybdo blue complex. In this assay, phosphate reacts in an acidic medium with ammonium molybdate to form a complex that, upon reduction, produces an intense blue coloration with a maximum absorption at 820 nm. The measurement of absorbed radiation (absorbance, A) is proportional to the concentration of the absorbing species, thereby allowing quantification.

A 10.0 mL aliquot of the **M₀** stock solution was taken and transferred to a 100 mL volumetric flask. The reagents necessary for color development were added, and the solution was flushed with distilled water (**M₂** Solution). The absorbance of a 25.0 mL aliquot of this dilute solution was measured, obtaining an A1 reading of 0.405. To a second aliquot of 25.0 mL of the same solution, 5.0 mL of a standard phosphate solution with a concentration of 0.200 mg/mL was added. The resulting mixture has an A2 absorbance of 0.667 under the same experimental conditions.

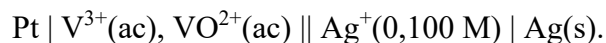
16.5. Calculate the phosphate concentration (in mg/mL) in the **M₂** solution. Consider that the Lambert-Beer Law, $A = k \cdot c(X)$, is true, with k being constant for both measures.

16.6. Determine the phosphorus content (in mg) in the initial cereal sample. If you were unable to determine the phosphate concentration in the **M₂** solution, use 0.0400 mg/mL in your calculations.

Part III. As a third method of quantification, phosphate was determined by quantitative precipitation as oxovanadium(IV) phosphate, $(VO)_3(PO_4)_2$. A 100 mL aliquot of the **M₀** solution was taken, slightly acidified and heated to 60 °C. Subsequently, an excess of 0.050 M standard $VOSO_4$ solution was slowly added to ensure complete precipitation. After filtration, washing and drying in the oven to constant mass, a precipitate mass reading of 80.4 mg was obtained.

16.7. Determine the phosphorus content (in mg) in the initial cereal sample.

Part IV. To determine the solubility product (K_{ps}) and molar solubility of the $(VO)_3(PO_4)_2$, a sufficient amount of the solid was brought into contact with a buffering solution of pH 6.0 acetate. A controlled amount of V^{3+} was added to this system, and the system was kept in agitation until equilibrium was reached. By ICP-OES analysis, the total concentration of vanadium ($V^{3+} + VO^{2+}$) dissolved in the supernatant was determined to be $1.0 \cdot 10^{-4}$ M. The following electrochemical cell was then constructed using the saturated solution of vanadium ions as the anodic electrolyte:



The measured cell potential was 1.128 V at 25 °C. $E^\circ(VO^{2+}/V^{3+}) = 0.337$ V, $E^\circ(Ag^+/Ag) = 0.799$ V.

16.8. Calculate the solubility product constant of $(VO)_3(PO_4)_2$.

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